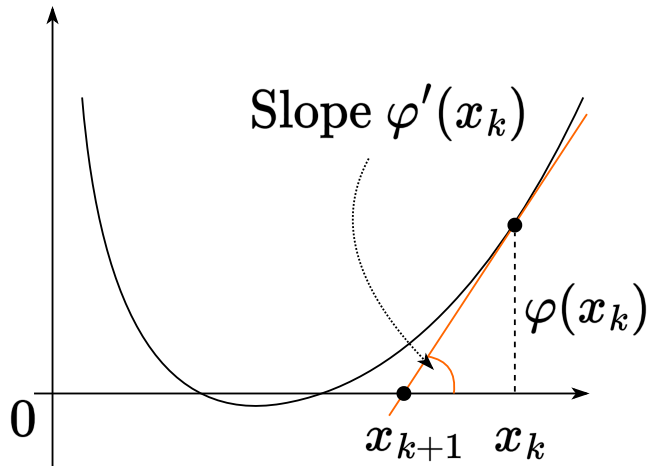


# Newton method. Quasi-Newton methods

## Seminar

Optimization for ML. Faculty of Computer Science. HSE University

## First interpretation of Newton method (solution of linearized equations)



Consider the function  $\varphi(x) : \mathbb{R} \rightarrow \mathbb{R}$ . We want to find the root of  $\varphi(x) = 0$ .

The whole idea came from building a linear approximation at the point  $x_k$  and find its root, which will be the new iteration point:

$$\varphi'(x_k) = \frac{\varphi(x_k)}{x_{k+1} - x_k}$$

We get an iterative scheme:

$$x_{k+1} = x_k - \frac{\varphi(x_k)}{\varphi'(x_k)}.$$

Now, if we consider  $\varphi(x) \equiv \nabla f(x)$ , this will become a Newton optimization method:

$$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$$

## Example of Newton linearization method

### Question

Apply Newton method to find the root of  $\varphi(t) = 0$  and determine the convergence area:

$$\varphi(t) = \frac{t}{\sqrt{1+t^2}}$$

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2. Then the iteration of the method takes the form:

$$x_{k+1} = x_k - \frac{\varphi(x_k)}{\varphi'(x_k)} = x_k - x_k(x_k^2 + 1) = -x_k^3$$

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It is easy to see that the method converges only if  $|x_0| < 1$ , emphasizing the **local** nature of the Newton method.

## Second interpretation of Newton method (local quadratic Taylor approximation minimizer)

Let us now have the function  $f(x)$  and a certain point  $x_k$ . Let us consider the quadratic approximation of this function near  $x_k$ :

$$f_{x_k}^{II}(x) = f(x_k) + \langle \nabla f(x_k), x - x_k \rangle + \frac{1}{2} \langle \nabla^2 f(x_k)(x - x_k), x - x_k \rangle.$$

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The idea of the method is to find the point  $x_{k+1}$ , that minimizes the function  $f_{x_k}^{II}(x)$ , i.e.  $\nabla f_{x_k}^{II}(x_{k+1}) = 0$ .

$$\nabla f_{x_k}^{II}(x_{k+1}) = \nabla f(x_k) + \nabla^2 f(x_k)(x_{k+1} - x_k) = 0$$

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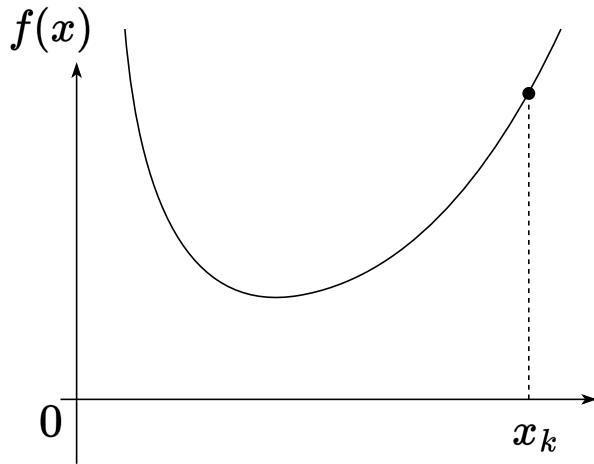
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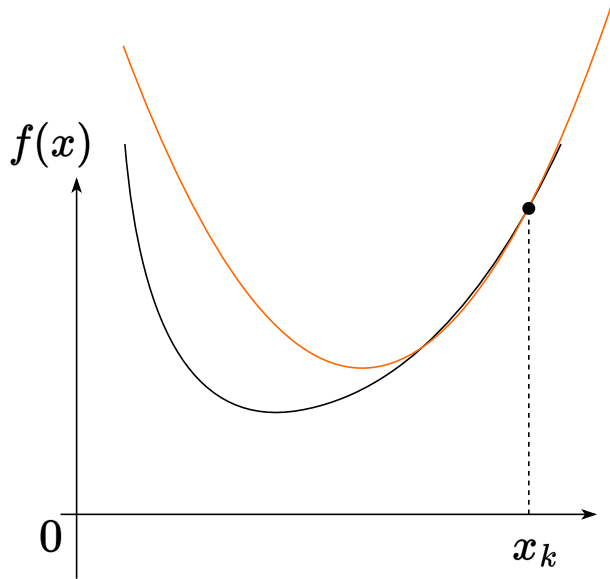
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Pay attention to the restrictions related to the need for the Hessian to be non-degenerate (for the method to work), as well as for it to be positive definite (for convergence guarantee).

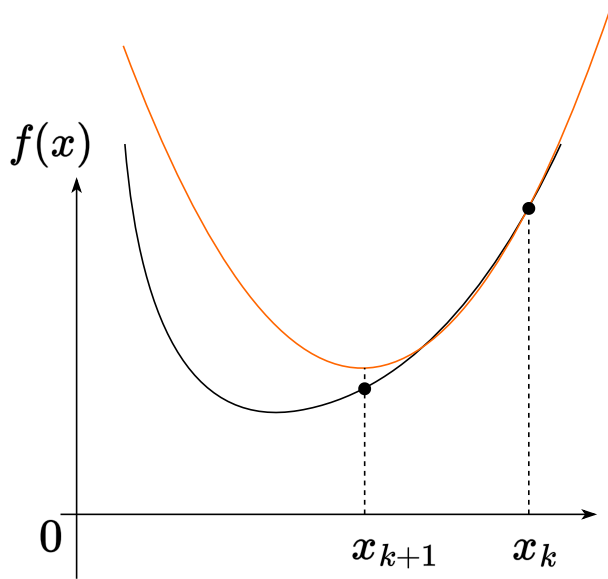
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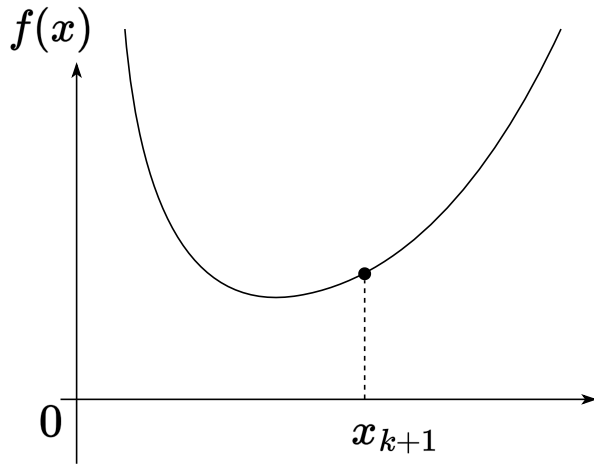
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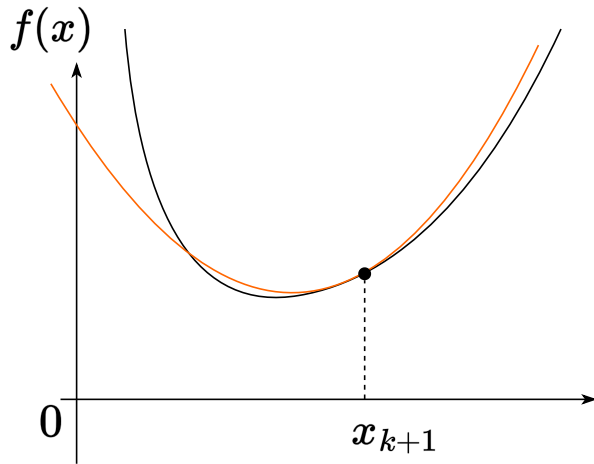
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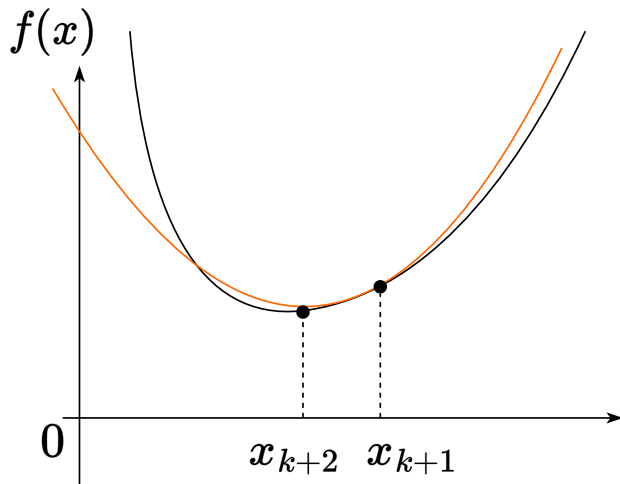
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# Newton method vs gradient descent

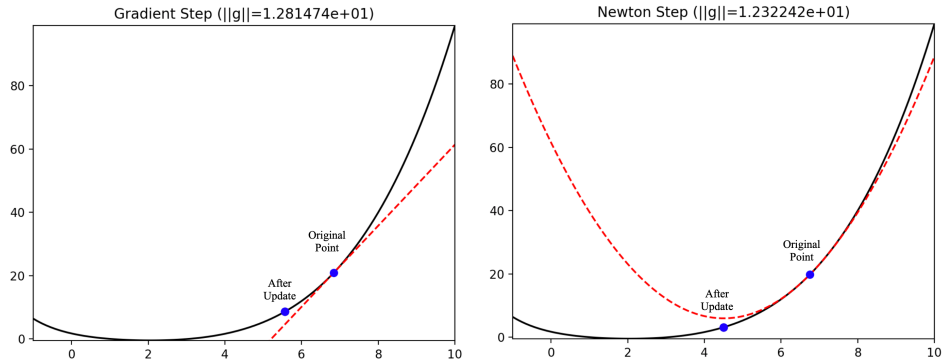


Figure 7: The loss function is depicted in black, the approximation as a dotted red line

The gradient descent  $\equiv$  linear approximation

The Newton method  $\equiv$  quadratic approximation



# Convergence

## i Theorem

Let  $f(x)$  be a strongly convex twice continuously differentiable function at  $\mathbb{R}^n$ , for the second derivative of which inequalities are executed:  $\mu I_n \preceq \nabla^2 f(x) \preceq L I_n$ . Then Newton method with a constant step

$$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$$

locally converges to solving the problem with superlinear speed. If, in addition, Hessian is  $M$ -Lipschitz continuous, then this method converges locally to  $x^*$  at a quadratic rate:

$$\|x_{k+1} - x^*\|_2 \leq \frac{M \|x_k - x^*\|_2^2}{2(\mu - M \|x_k - x^*\|_2)}$$

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**“Converge locally”** means that the convergence rate described above is guaranteed to occur only if the starting point is quite close to the minimum point, in particular  $\|x_0 - x^*\| < \frac{2\mu}{3M}$

## Affine invariance

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$$g(y+u) \approx g(y) + \langle g'(y), u \rangle + \frac{1}{2} u^\top g''(y) u \rightarrow \min_u$$

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4. Thus, the method's step is transformed by linear transformation in **the same way** as the coordinates:

$$Ay_{k+1} = Ay_k - (f''(Ay_k))^{-1} f'(Ay_k) \quad x_{k+1} = x_k - (f''(x_k))^{-1} f'(x_k)$$

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## Pros

- quadratic convergence near the solution
- high accuracy of the obtained solution
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- no global convergence
- it is necessary to store the hessian on each iteration:  $\mathcal{O}(n^2)$  memory
- it is necessary to solve linear systems:  $\mathcal{O}(n^3)$  operations
- the Hessian can be degenerate
- the Hessian may not be positively determined  $\rightarrow$  direction  $-(f''(x))^{-1}f'(x)$  may not be a descending direction 

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Cubic-regularized Newton method and Quasi Newton methods partially solve these problems!

# Intuition on how to improve Newton method

## 💡 Gradient Descent recap

If  $f$  has  $L$ -Lipschitz gradient, then

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

So, each step of gradient descent for function  $f$  with  $L$ -Lipschitz gradient is a minimization of majorizing paraboloid:

$$\begin{aligned} x_{k+1} &= \arg \min_x \left\{ f(x_k) + \langle \nabla f(x_k), x - x_k \rangle + \frac{L}{2} \|x - x_k\|^2 \right\} \\ &= x_k - \frac{1}{L} \nabla f(x_k). \end{aligned}$$

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But if function  $f$  has  $M$ -Lipschitz Hessian, it is easy to show that

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{1}{2} \langle \nabla^2 f(x)(y - x), y - x \rangle + \frac{M}{6} \|y - x\|^3.$$

**What if we use the same logic as in gradient descent for function with  $M$ -Lipschitz Hessian?**

# Cubic-regularized Newton method

If  $f$  has  $M$ -Lipschitz Hessian, then

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Minimizing the right-hand side of this inequality, we come to Cubic-regularized Newton method

$$x_{k+1} = \arg \min_x \left\{ f(x_k) + \langle \nabla f(x_k), x - x_k \rangle + \frac{1}{2} \langle \nabla^2 f(x_k)(x - x_k), x - x_k \rangle + \frac{M}{6} \|x - x_k\|^3 \right\}. \quad (1)$$

## Question

What problems do you see in (2)?

## Cubic-regularized Newton method

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### ! Challenges

1. We can't get explicit expression for  $x_{k+1}$  (without argmin) from (2) as we could in gradient descent.
2. The subproblem inside (2) can be non-convex.

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## 💡 Solutions

1. We can use numerical methods with fast convergence
2. The subproblem is equivalent to a convex one-dimensional optimization problem. <sup>a</sup>
3. The subproblem can be made convex with proper regularization coefficient. <sup>b</sup>

<sup>a</sup>Nesterov, Y. (2018). Lectures on convex optimization. Springer.

<sup>b</sup>Nesterov, Y. (2021). Implementable tensor methods in unconstrained convex optimization. Mathematical Programming.

## **i** Theorem

Let  $f(x)$  be  $\mu$ -strongly convex function with  $M$ -Lipschitz Hessian. Then, Cubic-regularized Newton Method (2) converges globally superlinearly as

$$f(x_{k+1}) - f^* \leq \gamma_k (f(x_k) - f^*), \quad \gamma_k \rightarrow 0.$$

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<sup>1</sup>Kamzolov, D., et al. (2024). Optami: Global superlinear convergence of high-order methods. Accepted to ICLR 2025.



# Quasi-Newton methods intuition

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Note here that if we take a single matrix of  $B_k = I_n$  as  $B_k$  at each step, we will exactly get the gradient descent method.

The general scheme of quasi-Newton methods is based on the selection of the  $B_k$  matrix so that it tends in some sense at  $k \rightarrow \infty$  to the truth value of the Hessian  $\nabla^2 f(x_k)$ .

# Quasi-Newton method Template

Let  $x_0 \in \mathbb{R}^n$ ,  $B_0 \succ 0$ . For  $k = 1, 2, 3, \dots$ , repeat:

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In addition to the secant equation, we want:

- $B_{k+1}$  to be symmetric
- $B_{k+1}$  to be “close” to  $B_k$
- $B_k \succ 0 \Rightarrow B_{k+1} \succ 0$

## Problem 1: Symmetric Rank-One (SR1) update

Let's try an update with rank-one matrix:

$$B_{k+1} = B_k + a u u^T$$

### Question

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# SR1 convergence

$$B_{k+1} = B_k + \frac{(\Delta y_k - B_k d_k)(\Delta y_k - B_k d_k)^T}{(\Delta y_k - B_k d_k)^T d_k}$$

called the symmetric rank-one (SR1) update or Broyden method.

## Theorem

Let

- $f$  be twice continuously differentiable, has unique stationary point  $x^*$ ,
- $0 \succ \nabla^2 f(x^*)$ ,  $\nabla^2 f(x)$  is Lipschitz continuous in a neighborhood  $x^*$ ,
- the sequence of matrices  $\{B_k\}$  is bounded in norm,
- $|(\Delta y_k - B_k d_k)^T d_k| \geq r \|d_k\| \|\Delta y_k - B_k d_k\|$ ,  $0 < r \ll 1$ .

Then in SR1  $x_k \rightarrow x^*$  superlinearly.

## SR1 with inverse update

How can we solve

$$B_{k+1}d_{k+1} = -\nabla f(x_{k+1}),$$

in order to take the next step? In addition to propagating  $B_k$  to  $B_{k+1}$ , let's propagate inverses, i.e.,  $C_k = B_k^{-1}$  to  $C_{k+1} = (B_{k+1})^{-1}$ .

### Sherman-Morrison Formula:

The Sherman-Morrison formula states:

$$(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1 + v^T A^{-1}u}$$

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Thus, for the SR1 update, the inverse is also easily updated:

$$C_{k+1} = C_k + \frac{(d_k - C_k \Delta y_k)(d_k - C_k \Delta y_k)^T}{(d_k - C_k \Delta y_k)^T \Delta y_k}$$

In general, SR1 is simple and cheap, but it has a key drawback: it does not preserve positive definiteness.

## Problem 2: Broyden-Fletcher-Goldfarb-Shanno (BFGS) update

Let's now try a rank-two update:

$$B_{k+1} = B_k + a u u^T + b v v^T.$$

### Question

What  $a$ ,  $u$ ,  $b$  and  $v$  can we choose? How the update of the  $B_{k+1}$  would look like?

## BFGS convergence

$$B_{k+1} = B_k - \frac{B_k d_k d_k^T B_k}{d_k^T B_k d_k} + \frac{\Delta y_k \Delta y_k^T}{d_k^T \Delta y_k}$$

called the Broyden-Fletcher-Goldfarb-Shanno (BFGS) update.

### Theorem

Let  $f(x)$  be twice continuously differentiable, have Lipschitz Hessian and additionally  $\sum_{k=1}^{\infty} \|x_k - x^*\| \leq \infty$ . Then in BFGS  $x_k \rightarrow x^*$  superlinearly.



# BFGS update with inverse

## Woodbury Formula

The Woodbury formula, a generalization of the Sherman-Morrison formula, is given by:

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$$

# BFGS update with inverse

## Woodbury Formula

The Woodbury formula, a generalization of the Sherman-Morrison formula, is given by:

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$$

Applied to our case, we get a rank-two update on the inverse  $C$ :

$$C_{k+1} = C_k + \frac{(d_k - C_k \Delta y_k) d_k^T}{\Delta y_k^T d_k} + \frac{d_k (d_k - C_k \Delta y_k)^T}{\Delta y_k^T d_k} - \frac{(d_k - C_k \Delta y_k)^T \Delta y_k}{(\Delta y_k^T d_k)^2} d_k d_k^T$$

$$C_{k+1} = \left( I - \frac{d_k \Delta y_k^T}{\Delta y_k^T d_k} \right) C_k \left( I - \frac{\Delta y_k d_k^T}{\Delta y_k^T d_k} \right) + \frac{d_k d_k^T}{\Delta y_k^T d_k}$$

This formulation ensures that the BFGS update, while comprehensive, remains computationally efficient, requiring  $O(n^2)$  operations. Importantly, BFGS update preserves positive definiteness. Recall this means  $B_k \succ 0 \Rightarrow B_{k+1} \succ 0$ . Equivalently,  $C_k \succ 0 \Rightarrow C_{k+1} \succ 0$

# Computational experiments

Let's look at computational experiments for Newton and Quasi Newton methods 🔗.